

Week 10-1 & 10-2

Lecture Note

Inventory – Price
(재고자산 – 가격)

Principles of Accounting
Dr. Joe (Joong Hi) Cho

Flow and Distribution of Inventory Costs

✓ Sales: COGS = Purchase Cost per Unit X Units Sold

✓ 판매분: 매출원가 = 단위당 매입원가 X 판매수량

2

1

✓ Not Sales: Ending Inventory = Purchase Cost per Unit X Units in Inventory

✓ 미판매분: 기말재고 = 단위당 매입원가 X 재고수량

1. Book sales quantity vs Actual inventory quantity?

장부상 판매수량을 기준으로 할 것인가, 실제 재고 수량을 기준으로 할 것인가?

2. If inventory (merchandise) is purchased at different prices over multiple occasions, what is the unit purchase cost?

재고자산(상품)을 여러 회에 걸쳐 다른 가격으로 매입한다면, 단위당 매입원가는?

Inventory Cost Flow Methods

Specific unit

Average cost

First-in, first-out

~~Last in, first out~~

Specific Identification Method (개별법)

Specific Identification Method (개별법)

- Used for business with unique inventory items (e.g., Automobiles, fine jewelry, real estate)
- Inventory costed at specific price of the particular unit
- Suitable for low transaction volume and high priced items
- Non-interchangeable inventory assets
- Goods or services produced for specific projects

Specific Identification Method (개별법)

Illustration: Crivitz TV Company purchases three identical 50-inch TVs on different dates at costs of £720, £750, and £800. During the year, Crivitz sold two sets at £1,200 each. These facts are summarized below.

Purchases			
February 3	1 TV	at	£720
March 5	1 TV	at	£750
May 22	1 TV	at	£800
Sales			
June 1	2 TVs	for	£2,400 (£1,200 × 2)

Illustration 6.3: Data for inventory costing example

Specific Identification Method (개별법)

If Crivitz sold the TVs it purchased on February 3 and May 22, then its cost of goods sold is £1,520 (£720 + £800), and its ending inventory is £750.

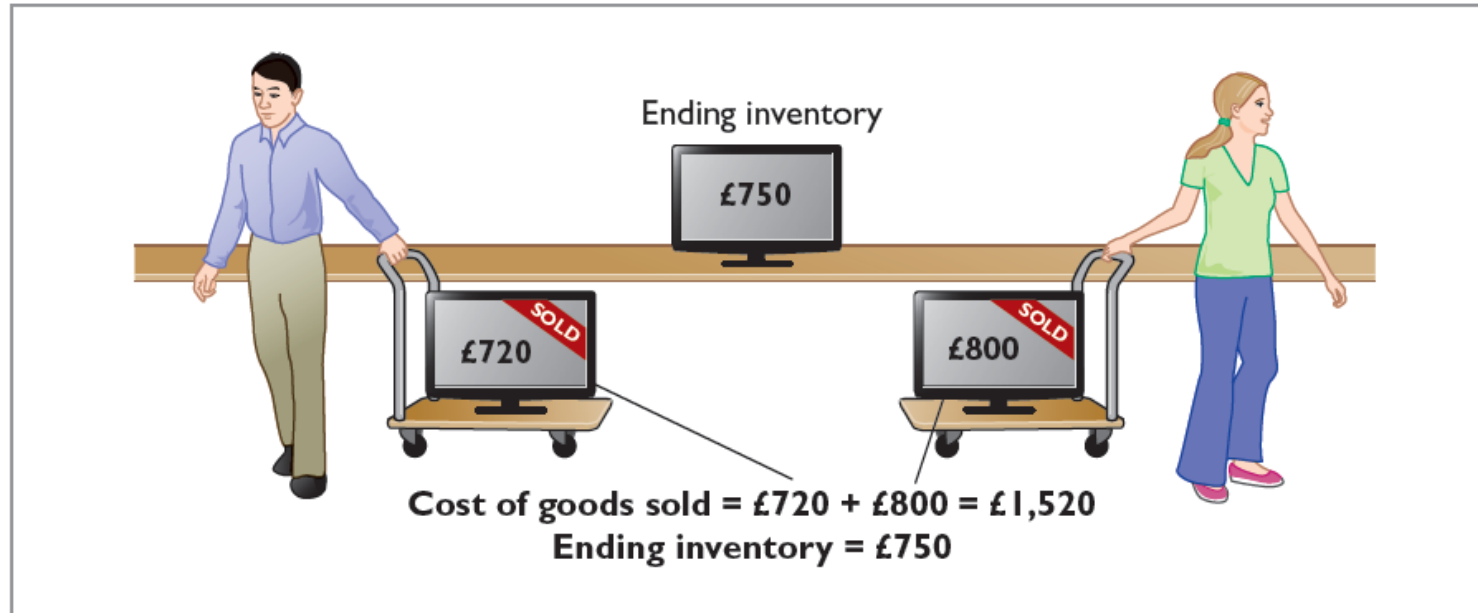
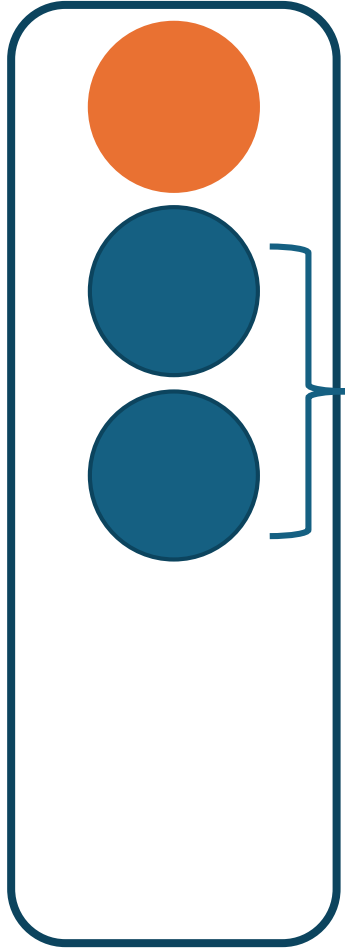


Illustration 6.4: Specific identification method

Specific Identification Method (개별법)

- Perpetual Inventory System (계속기록법)
- Periodic Inventory System (실지재고조사법)

Specific Units in the Perpetual Inventory System



1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

Inventory	
1/1 Beg. 100	
3/1 Cash 400	

Specific Units in the Perpetual Inventory System

1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

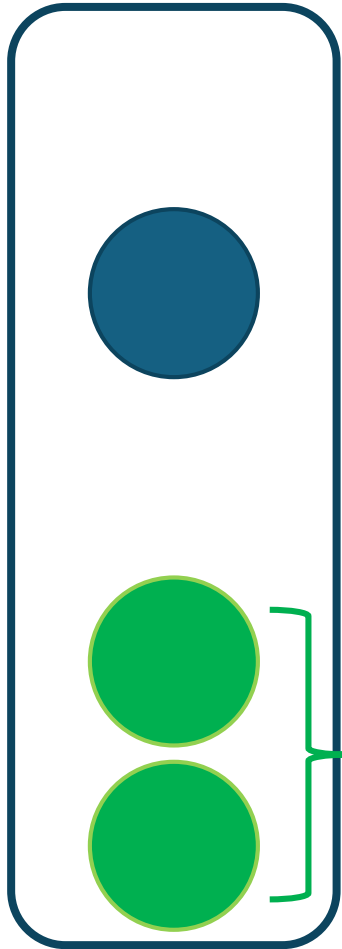
3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)

→ Dr. Cash 800 // Cr. Revenue 800
Dr. COGS 300 // Cr. Inventory 300

Inventory	
1/1 Beg. 100	6/1 COGS 300
3/1 Cash 400	

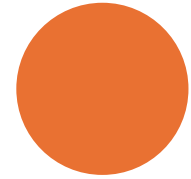
Specific Units in the Perpetual Inventory System



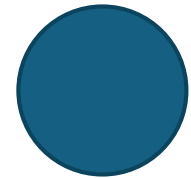
1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

8/1 Purchased 2 units at ₩300 each
(8/1 2개 매입, 단위당 ₩300)
→ Dr. Inventory 600 // Cr. Cash 600



6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)

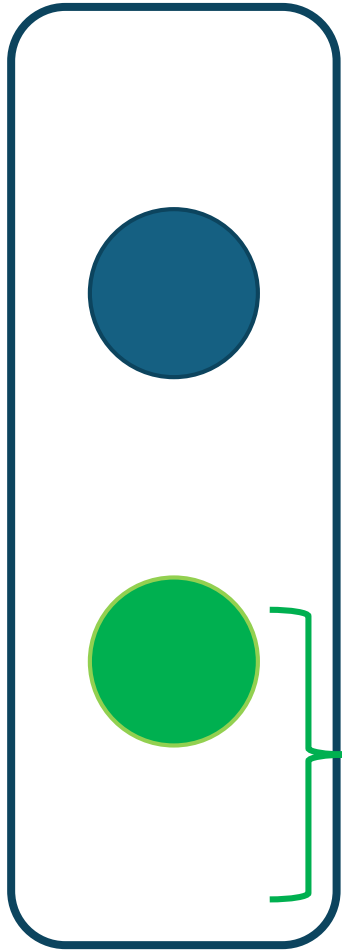


→ Dr. Cash 800 // Cr. Revenue 800
Dr. COGS 300 // Cr. Inventory 300

Inventory

1/1 Beg. 100	6/1 COGS 300
3/1 Cash 400	
8/1 Cash 600	

Specific Units in the Perpetual Inventory System



1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

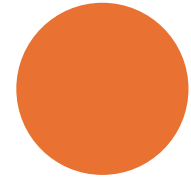
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(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

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(8/1 2개 매입, 단위당 ₩300)
→ Dr. Inventory 600 // Cr. Cash 600

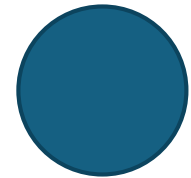
No other transactions until 12/31

Inventory

1/1 Beg. 100	6/1 COGS 300
3/1 Cash 400	9/1 COGS 300
8/1 Cash 600	End. Invt 500



6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)



→ Dr. Cash 800 // Cr. Revenue 800
Dr. COGS 300 // Cr. Inventory 300

9/1 Sell 1 units at ₩400
(9/1 1개 판매, 단위당 ₩400)



→ Cash 400 // Revenue 400
COGS 300 // Inventory 300

Specific Units in the Perpetual Inventory System

Income Statement (손익계산서)	
Sales 매출	₩ 400 X 3개 = ₩ 1,200
Cost of Goods Sold (COGS) 매출원가	₩ 600 (=100+200+300)
Gross Profit 매출총이익	₩ 600

Balance Sheet (재무상태표)	
Cash 현금	₩ XXX
Account Receivables 매출채권	₩ XXX
Inventory 상품	₩ 500 (200+300)

Specific Units in the Perpetual Inventory System

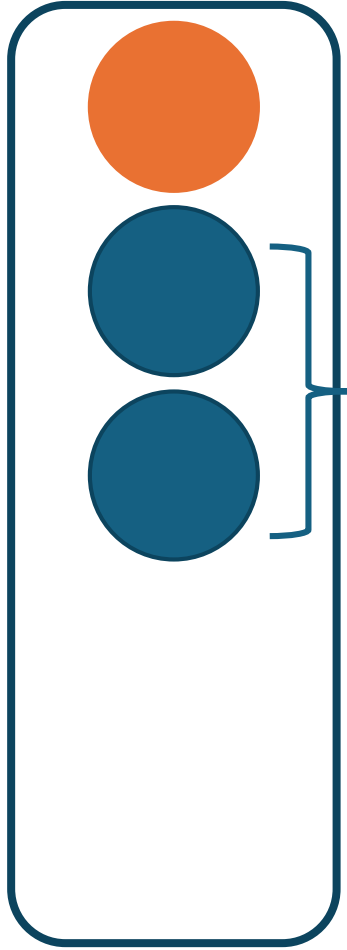
Income Statement (손익계산서)	
Sales 매출	₩ 400 X 3개 = ₩ 1,200
Cost of Goods Sold (COGS) 매출원가	₩ 600 (=100+200+300)
Gross Profit 매출총이익	₩ 600
Balance Sheet (재무상태표)	
Cash 현금	₩ XXX
Account Receivables 매출채권	₩ XXX
Inventory 상품	₩ 500 (200+300)

Sold units

Cost of Goods Available
for Sale ~~₩~~1,100

Not sold

Specific Units in the Periodic Inventory System



1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

→ Dr. **Purchase** 400 // Cr. Cash 400

Inventory(상품)	
1/1 Beg. 100	
Purchase(매입)	
3/1 Cash 400	

Specific Units in the Periodic Inventory System

1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

→ Dr. **Purchase** 400 // Cr. Cash 400

6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)

→ Dr. Cash 800 // Cr. Revenue 800

~~Dr. COGS 300 // Cr. Inventory 300~~

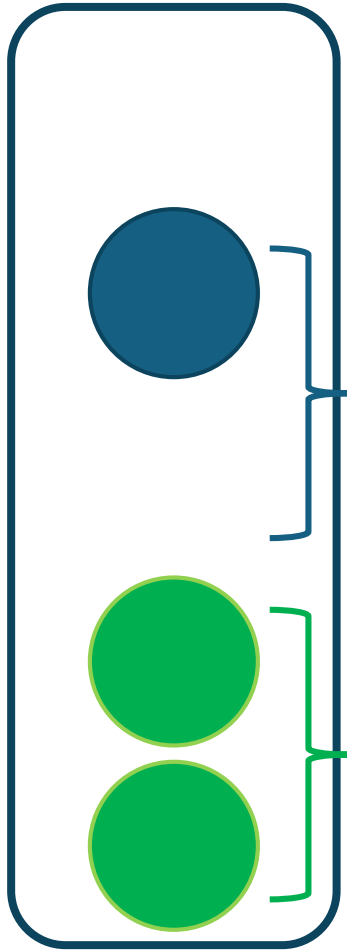
Inventory(상품)

1/1 Beg. 100

Purchase(매입)

3/1 Cash 400

Specific Units in the Periodic Inventory System



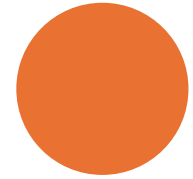
1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

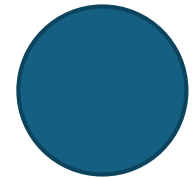
→ Dr. **Purchase** 400 // Cr. Cash 400

8/1 Purchased 2 units at ₩300 each
(8/1 2개 매입, 단위당 ₩300)

→ Dr. **Purchase** 600 // Cr. Cash 600



6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)



→ Dr. Cash 800 // Cr. Revenue 800

~~Dr. COGS 300 // Cr. Inventory 300~~

Purchase(매입)

3/1 Cash 400
8/1 Cash 600

Specific Units in the Periodic Inventory System

₩200

1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

→ Dr. **Purchase** 400 // Cr. Cash 400

₩300

8/1 Purchased 2 units at ₩300 each
(8/1 2개 매입, 단위당 ₩300)

→ Dr. **Purchase** 600 // Cr. Cash 600

12/31 Ending Inventory
(Physical count): 2 units

Purchase(매입)

3/1 Cash 400

8/1 Cash 600

6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)

→ Dr. Cash 800 // Cr. Revenue 800

~~Dr. COGS 300 // Cr. Inventory 300~~

9/1 Sell 1 units at ₩400
(9/1 1개 판매, 단위당 ₩400)

→ Cash 400 // Revenue 400

~~COGS 300 // Inventory 300~~

Specific Units in the Periodic Inventory System

Inventory

1/1 Beg Inventory	100	(1) 12/31 Beg Inventory	100
(2) 12/31 End Inventory	500		
(physical check)			

COGS

(3) 12/31 COGS	600	(4) 12/31 COGS	600
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Purchase

3/1 Cash	400		
8/1 Cash	600		
(1) 12/31 Beg Inventory	100	(2) 12/31 End Inventory	500
		(physical check)	
Cost of Goods Available for Sale		(3) 12/31 COGS	600 [Plug-In]
1,100			

Income Summary

(4) 12/31 COGS	600
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Specific Units in the Periodic Inventory System

Income Statement (손익계산서)	
Sales 매출	$\cancel{\text{₩}}400 \times 3\text{개} = \cancel{\text{₩}}1,200$
Cost of Goods Sold (COGS) 매출원가 $\cancel{\text{₩}}100 \text{ (Beg)} + \cancel{\text{₩}}1,000 \text{ (Purchased)} - \cancel{\text{₩}}500 \text{ (End)}$	Sold units $\cancel{\text{₩}}600$
Gross Profit 매출총이익	$\cancel{\text{₩}}600$
Cost of Goods Available for Sale $\cancel{\text{₩}}1,100$	
Balance Sheet (재무상태표)	
Cash 현금	$\cancel{\text{₩}}XXX$
Account Receivables 매출채권	Not sold $\cancel{\text{₩}}XXX$
Inventory 상품	$\cancel{\text{₩}}500$

Physically count first at the end of the accounting period

Specific Identification Method (개별법)

- Costing method in which items still in inventory are specifically costed to arrive at the total cost of the ending inventory
(재고에 남아 있는 물건들이 각각 얼마에 샀는지를 정확히 추적해서, 기말재고 총액을 계산하는 방식)
 - Exact matching between cost and revenue
 - Useful for high-value, unique items

However,

- Practice is relatively rare (실제로는 드물게 사용)
- Not practical for mass inventory
- High administrative effort
- Potential for earnings manipulation

Specific Identification Method (개별법)

Thus, in practical,

- Most companies make assumptions (cost flow assumptions) about which units were sold (대부분의 기업은 어떤 상품이 먼저 팔렸는지에 대해 '가정(원가흐름 가정)'을 세워서 처리)
→ First-in, first-out (FIFO) and Average-cost

First-in, First-out (FIFO)

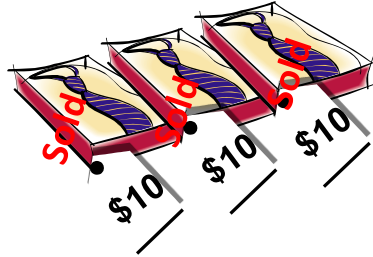
선입선출법

First-in, First-out (FIFO)

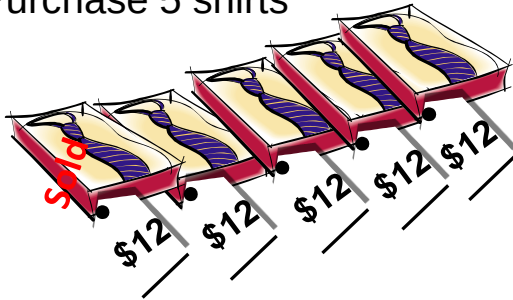
- Assumes the oldest inventory is sold first, so ending inventory reflects the most recent costs.
- Common and intuitive method, especially when inventory turnover follows a natural flow (e.g., food, perishable goods).

First-in, First-out (FIFO)

Beginning Inventory: 3 shirts



Purchase 5 shirts



We sell 4 shirts for \$20 each.
What costs should be assigned to
Cost of Goods Sold? To inventory?

First-In, First-Out

$$\text{Cost of good sold} = 10 \times 3 + 12 \times 1 = 42$$

Sell the oldest ones first

$$\text{Ending Inventory} = 12 \times 4 = 48$$

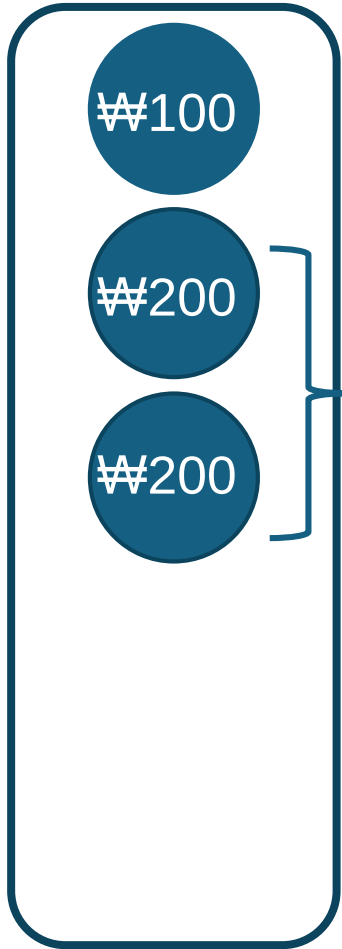
Keep the most recent ones

First-in, First-out (FIFO)

Feb 11	Accounts Receivable (\$20 x 4)	80	
	Sales Revenue		80
	<i>Sales on account</i>		

Feb 11	Cost of Goods Sold	42	
	Inventory		42
	<i>Cost of sales</i>		

FIFO under the Perpetual Inventory System

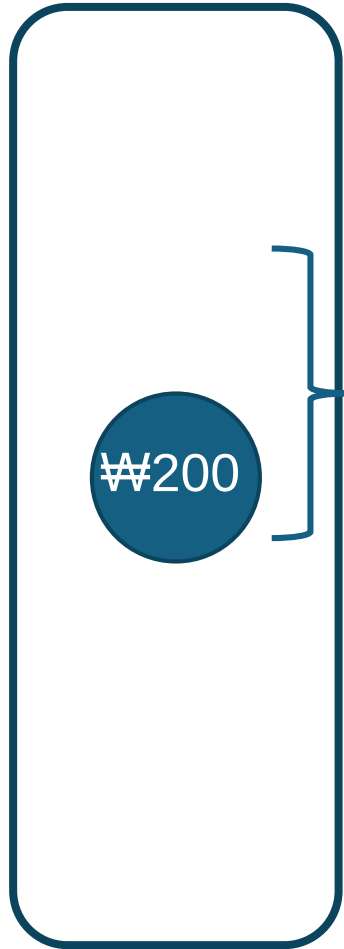


1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

Inventory	
1/1 Beg. 100	
3/1 Cash 400	

FIFO under the Perpetual Inventory System



1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

₩100

₩200

6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)
→ Dr. Cash 800 // Cr. Revenue 800
Dr. COGS 300 // Cr. Inventory 300

Inventory	
1/1 Beg. 100	6/1 COGS 300
3/1 Cash 400	

FIFO under the Perpetual Inventory System

1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

₩100

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)
→ Dr. Inventory 400 // Cr. Cash 400

₩200

6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)

→ Dr. Cash 800 // Cr. Revenue 800
Dr. COGS 300 // Cr. Inventory 300

8/1 Purchased 2 units at ₩300 each
(8/1 2개 매입, 단위당 ₩300)
→ Dr. Inventory 600 // Cr. Cash 600

₩200

9/1 Sell 1 units at ₩400
(9/1 1개 판매, 단위당 ₩400)
→ Cash 400 // Revenue 400
COGS 200 // Inventory 200

No other transactions until 12/31

Inventory

1/1 Beg. 100	6/1 COGS 300
3/1 Cash 400	9/1 COGS 200
8/1 Cash 600	End. Invt 600

FIFO under the Perpetual Inventory System

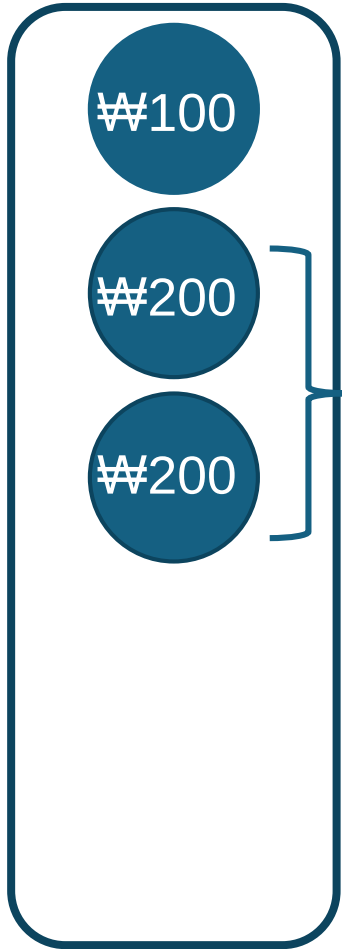
Income Statement (손익계산서)	
Sales 매출	₩ 400 X 3개 = ₩ 1,200
Cost of Goods Sold (COGS) 매출원가	₩ 500 (=100+200+200)
Gross Profit 매출총이익	₩ 700

Balance Sheet (재무상태표)	
Inventory 상품	₩ 600 (300+300) or (100+1,000-500)

FIFO under the Perpetual Inventory System

Income Statement (손익계산서)	
Sales 매출	$\cancel{\text{₩}}400 \times 3\text{개} = \cancel{\text{₩}}1,200$
Cost of Goods Sold (COGS) 매출원가	$\cancel{\text{₩}}500 (=100+200+200)$
Gross Profit 매출총이익	$\cancel{\text{₩}}700$
Balance Sheet (재무상태표)	
Inventory 상품	Cost of Goods Available for Sale $\cancel{\text{₩}}1,100$ Not sold $\cancel{\text{₩}}600$ (300+300) or (100+1,000-500)

FIFO under the Periodic Inventory System



1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

→ Dr. **Purchase** 400 // Cr. Cash 400

Inventory(상품)	
1/1 Beg. 100	
Purchase(매입)	
3/1 Cash 400	

FIFO under the Periodic Inventory System

1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

₩100

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

₩200

→ Dr. **Purchase** 400 // Cr. Cash 400

6/1 Sell 2 units at ₩400

(6/1 2개 판매, 단위당 ₩400)

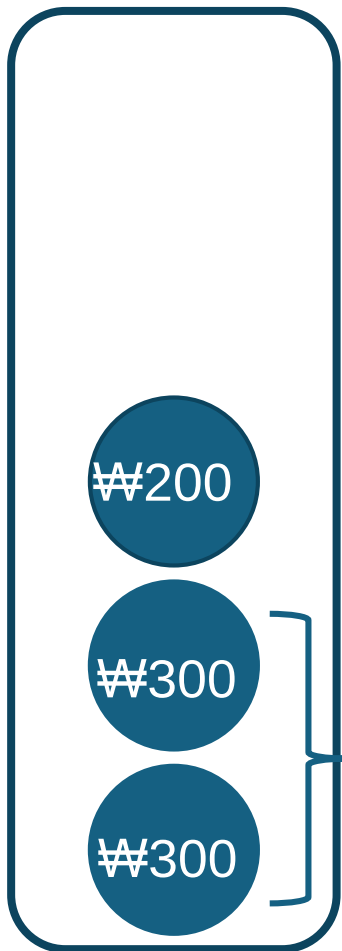
→ Dr. Cash 800 // Cr. Revenue 800

~~Dr. COGS 300 // Cr. Inventory 300~~

Purchase(매입)

3/1 Cash 400

FIFO under the Periodic Inventory System



1/1 Beginning Inventory 1 unit at ₩100 each
(1/1 기초 1개, 단위당 ₩100)

3/1 Purchased 2 units at ₩200 each
(3/1 2개 매입, 단위당 ₩200)

→ Dr. **Purchase** 400 // Cr. Cash 400

8/1 Purchased 2 units at ₩300 each
(8/1 2개 매입, 단위당 ₩300)

→ Dr. **Purchase** 600 // Cr. Cash 600

12/31 Ending Inventory
(Physical count): 2 units (₩600)

Purchase(매입)	
3/1 Cash 400	
8/1 Cash 600	

₩100

₩200

6/1 Sell 2 units at ₩400
(6/1 2개 판매, 단위당 ₩400)

→ Dr. Cash 800 // Cr. Revenue 800

~~Dr. COGS 300 // Cr. Inventory 300~~

₩200

9/1 Sell 1 units at ₩400
(9/1 1개 판매, 단위당 ₩400)

→ Cash 400 // Revenue 400

~~COGS 200 // Inventory 200~~

FIFO under the Periodic Inventory System

Inventory

1/1 Beg Inventory	100	(1) 12/31 Beg Inventory	100
(2) 12/31 End Inventory	600		
(physical check)			

COGS

(3) 12/31 COGS	500	(4) 12/31 COGS	500
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Purchase

3/1 Cash	400		
8/1 Cash	600		
(1) 12/31 Beg Inventory	100	(2) 12/31 End Inventory	600
		(physical check)	
Cost of Goods Available for Sale		(3) 12/31 COGS	500 [Plug-In]
1,100			

Income Summary

(4) 12/31 COGS	500
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Specific Units in the Periodic Inventory System

Income Statement (손익계산서)	
Sales 매출	$\cancel{\text{₩}}400 \times 3\text{개} = \cancel{\text{₩}}1,200$
Cost of Goods Sold (COGS) 매출원가 $\cancel{\text{₩}}100 \text{ (Beg)} + \cancel{\text{₩}}1,000 \text{ (Purchased)} - \cancel{\text{₩}}600 \text{ (End)}$	Sold units $\cancel{\text{₩}}500$
Gross Profit 매출총이익	$\cancel{\text{₩}}700$
Cost of Goods Available for Sale $\cancel{\text{₩}}1,100$	
Balance Sheet (재무상태표)	
Cash 현금	$\cancel{\text{₩}}\text{XXX}$
Account Receivables 매출채권	Not sold $\cancel{\text{₩}}\text{XXX}$
Inventory 상품	$\cancel{\text{₩}}600$

Physically count first at the end of the accounting period

First-in, First-out (FIFO)

- Matching the normal flow of goods → Reasonable
 - First purchased items are sold first
 - Easy to apply in most industries
- Cost-Profit mismatch (shortcoming)
 - In periods of rising prices:
 - Cost of Goods Sold (COGS) is understated
 - Net income is overstated
 - Higher tax burden
 - Increased dividend pressure on shareholders
- Ending inventory reflects recent prices
 - More accurately represents current inventory value
- Applicable under both inventory system (perpetual and periodic)

Average Cost

평균법

Average Cost Method

- Assigns the same average unit cost to all items, regardless of purchase date.
- Simple to apply and smooths out price fluctuations in cost of goods sold.
- Allocates cost of goods available for sale on the basis of weighted-average unit cost incurred.
- Applies weighted-average unit cost to units on hand to determine cost of ending inventory → Thus, the Average Cost Method only can be used under the Periodic Inventory System.
- Assumes that goods are similar in nature.

Average Cost Method

$$\text{Average cost per unit} = \frac{\text{Cost of goods available}^*}{\text{Number of units available}^*}$$

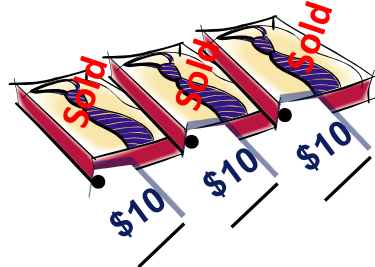
*Goods available = Beginning inventory + Purchases

$$\text{Cost of goods sold} = \text{Number of units sold} \times \text{Average cost per unit}$$

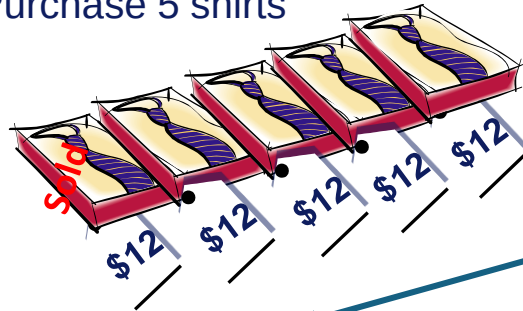
$$\text{Ending inventory} = \text{Number of units on hand} \times \text{Average cost per unit}$$

Average Cost Method

Beginning Inventory



Purchase 5 shirts



- Beg.Invt 3 + Purchased 5 = total 8 shirts (available for sale)
- We sell 4 shirts → 4 shirts are left

Compute the Average Cost

	Units	Cost
Beginning inventory	3	\$30
Purchases	<u>5</u>	<u>60</u>
Total	8	\$90

Average = $\$90/8 = \11.25

Inventory = $\$11.25 \times 4 = \45

Cost of good sold = $\$11.25 \times 4 = \45

Using Inventory Cost Flow Methods Consistently

- The method chosen should be used **consistently**, enhancing comparability over time
- Although consistency is preferred, a company may change its inventory costing method
 - Should disclose the change and its effect on net income

Average Cost Method

$$\text{Average cost per unit} = \frac{\text{Cost of goods available}^*}{\text{Number of units available}^*}$$

*Goods available = Beginning inventory + Purchases

$$\text{Cost of goods sold} = \text{Number of units sold} \times \text{Average cost per unit}$$

$$\text{Ending inventory} = \text{Number of units on hand} \times \text{Average cost per unit}$$

↓
**Physical count ending inventory
(i.e., the periodic inventory system)**

Question1

The cost flow method that often parallels the actual physical flow of merchandise is the:

- a. FIFO method
- b. LIFO method
- c. average cost method
- d. gross profit method

Solution

The cost flow method that often parallels the actual physical flow of merchandise is the:

- a. FIFO method (correct)**
- b. LIFO method
- c. average cost method
- d. gross profit method

Question2

The accounting records of Shumway Implements show the following data.

Beginning inventory	4,000 units at € 3
Purchases	6,000 units at € 4
Sales	7,000 units at €12

Determine the cost of goods sold during the period under a periodic inventory system using (a) the FIFO method, and (b) the average-cost method.

Solution

Cost of goods available for sale = $(4,000 \times \text{€}3) + (6,000 \times \text{€}4) = \text{€}36,000$

Ending inventory = $10,000 - 7,000 = 3,000$ units

a. FIFO: $\text{€}36,000 - (3,000 \times \text{€}4) = \text{€}24,000$

b. Average cost per unit: $[(4,000 @ \text{€}3) + (6,000 @ \text{€}4)] \div 10,000 = \text{€}3.60$

Average-cost: $\text{€}36,000 - (3,000 \times \text{€}3.60) = \text{€}25,200$

Inventory Turnover

Financial Analysis and Data Analytics

Inventory management is a double-edged sword

- 1. High Inventory Levels** – may incur high carrying costs (e.g., investment, storage, insurance, obsolescence, and damage)
- 2. Low Inventory Levels** – may lead to stock-outs and lost sales

Analysis Measure

Inventory turnover measures the number of times on average the inventory is sold during the period.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Days in inventory measures the average number of days inventory is held.

$$\text{Days in Inventory} = \frac{\text{Days in Year (365)}}{\text{Inventory Turnover}}$$

Analysis Example

Illustration: Esprit Holdings (HKG) reported in a recent annual report a beginning inventory of HK\$3,209 million, an ending inventory of HK\$3,254 million, and cost of goods sold for the year ended of HK\$12,071 million. The illustration shows the **inventory turnover** formula and computation for Esprit Holdings.

$$\begin{array}{rclcl} \text{Cost of} & & \text{Average} & & \\ \text{Goods Sold} & \div & \text{Inventory} & = & \text{Inventory Turnover} \\ \\ \text{HK\$12,071} & \div & \frac{\text{HK\$3,209} + \text{HK\$3,254}}{2} & = & 3.7 \text{ times} \end{array}$$

Illustration 6.16: Inventory turnover formula and computation for Esprit Holdings (in millions)

Data Analytics

- Companies can use data analytics to improve inventory management.
- Example: Walmart (USA) collects data about every inventory item and customer. They use it to analyze customer habits, buying patterns and sales trends.

Question3

Early in 2025, Westmoreland Company switched to a just-in-time inventory system. Its sales revenue, cost of goods sold, and inventory amounts for 2024 and 2025 are shown below.

	2024	2025
Sales revenue	NT\$2,000,000	NT\$1,800,000
Cost of goods sold	1,000,000	910,000
Beginning inventory	290,000	210,000
Ending inventory	210,000	50,000

Determine the inventory turnover and days in inventory for 2024 and 2025.

Solution

	2024	2025
Inventory Turnover	$\frac{\text{NT\$1,000,000}}{(\text{NT\$290,000} + \text{NT\$210,000})/2} = 4$	$\frac{\text{NT\$910,000}}{(\text{NT\$210,000} + \text{NT\$50,000})/2} = 7$
Days in inventory	$365 \div 4 = 91.3 \text{ days}$	$365 \div 7 = 52.1 \text{ days}$

- The company experienced a very significant decline in its ending inventory as a result of the just-in-time inventory. This decline improved its inventory turnover and its days in inventory.
- Sales declined by 10%. May be caused by reduction in the amount of inventory that was on hand, which increased the “stockouts.” Management must weigh the benefits of reduced inventory against the potential lost sales caused by stock-outs.